Research Methodology-I
Quantitative Research

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A Quick Recap

- What do you understand by the term research?
- The creation of new knowledge and/or the use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings

- What do you understand by the term quantitative?
- Data represented numerically, including anything that can be counted, measured, or given a numerical value.

Objectives

At the end of the lecture student should be able to:

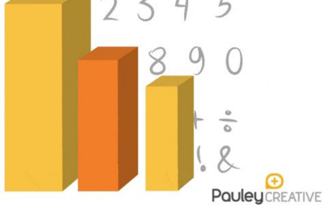
- Define quantitative research and explain its significance in health and dental sciences.
- Identify common types of quantitative research designs
- Understand with examples common types of quantitative research designs

Research

- A way of asking questions and finding answers using organized methods
- It involves the following basic steps:

- Identifying a problem or question – What do you want to find out?
- Reviewing what is already known – What have others said or discovered?
- Planning how to investigate – How will you collect information or data?
- Collecting and analyzing data – Gathering evidence and making sense of it.
- Drawing conclusions – What do your findings mean?
- Sharing results – Telling others what you discovered.

Quantitative Research Attributes



- Evidence-Based Practice:

Quantitative research provides strong, measurable evidence that helps dentists and doctors make scientific decisions about treatments.

- Improving Patient Care:

By analyzing data (e.g. how many patients respond to a treatment), we can improve techniques and outcomes in dental care.

- Understanding Disease Patterns:

It helps us study how common a disease is, what causes it, and how it spreads — very useful in epidemiology and public health.

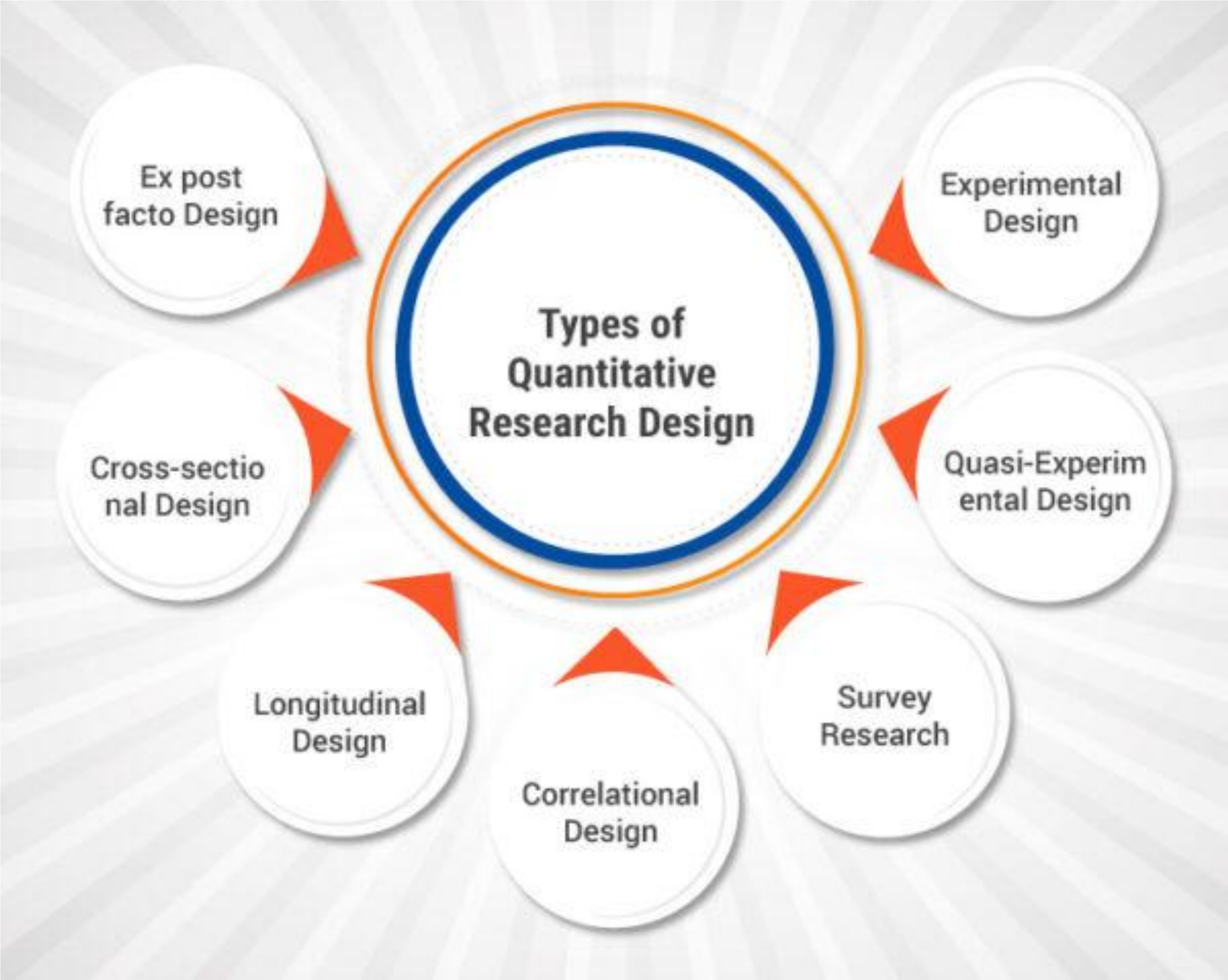
- Comparing Treatments:

We can use quantitative methods to compare different dental materials or procedures, and choose what works best.

- Standardization:

It allows us to create guidelines and protocols based on solid data, not just opinion or guesswork.

Quantitative Research: Types



Quantitative Research: Ex post facto

- Ex post facto (Latin for "after the fact") research is a type of non-experimental quantitative research where the researcher studies the effect of an event or variable after it has already occurred, without manipulating any variables.
- It looks backward to find possible causes of an observed effect.
- The researcher cannot control or change the variables — only observe and analyze them.
- This design is useful when random assignment or manipulation of variables is not feasible or ethical.

Quantitative Research: Ex post facto

- Example??????

A researcher wants to study whether prolonged use of carbonated drinks during teenage years is linked to increased enamel erosion in young adults.
The researcher cannot make students drink soda (for ethical reasons).Instead, they select individuals who already have enamel erosion and then analyze their past habits (e.g., frequency of soda intake).
This is ex post facto research because the researcher is studying existing effects (enamel erosion) and tracing back to find possible causes (soda consumption).

Quantitative Research: Cross-Section

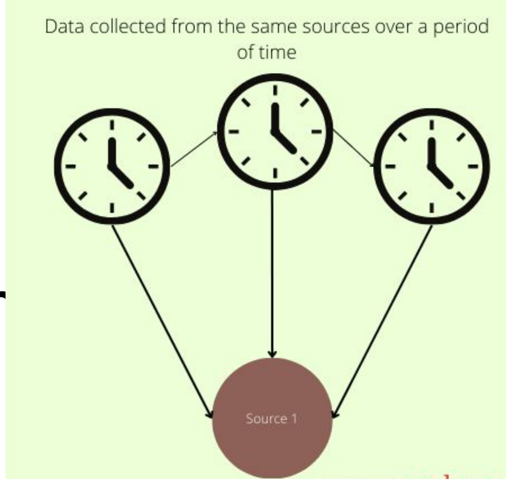
- Cross-sectional research is a type of observationa quantitative study where data is collected from a population at one specific point in time.
- It provides a "snapshot" of what is happening in a group right now.
- It helps describe the **prevalence** of health conditions or behaviors.
- Often uses surveys or clinical exams.
- Useful for identifying trends, needs, or risk factors.

Quantitative Research: Cross-Section

- Example??????

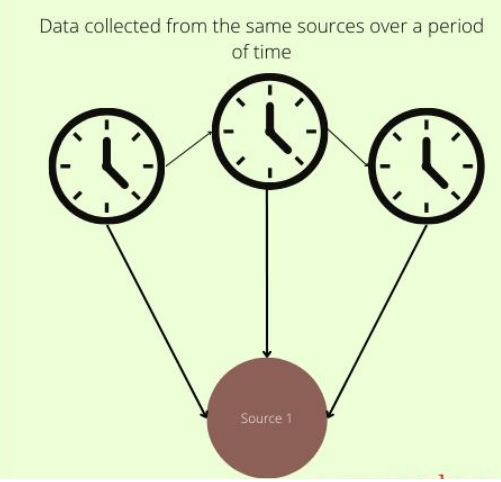
A dental researcher wants to find out how many 12-year-old children in a city have dental caries. They select a sample of 12-year-old students from different schools. They examine them once and record who has caries and who doesn’t. This is cross-sectional research because the data is collected at a single point in time, and no follow-up is involved.

Quantitative Research: Longitudinal



- A longitudinal study design involves collecting data from the same participants over an extended period of time.
- This design allows researchers to track changes in variables or outcomes over time and identify patterns or trends that emerge.
- It is particularly useful for studying developmental changes, impacts of interventions, or long-term effects of certain conditions or behaviors.
- Can provide stronger evidence of cause-and-effect relationships than cross-sectional studies.

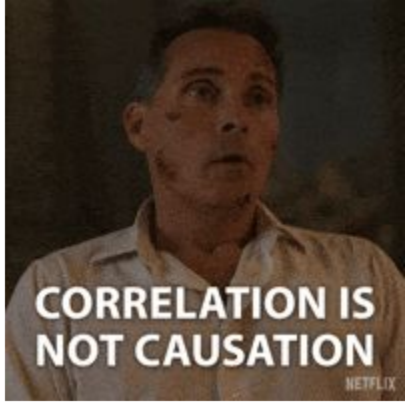
Quantitative Research: Longitudinal



- Example??????

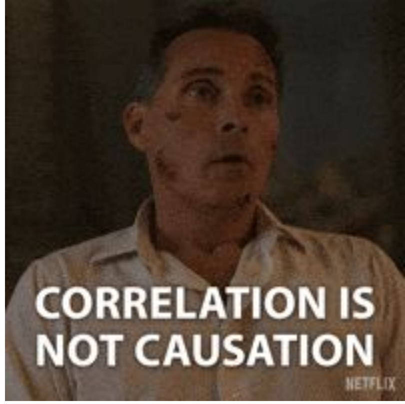
A dental researcher wants to study how orthodontic treatment affects oral hygiene in teenagers over 3 years.
A group of teenagers starting braces is selected.Their oral hygiene is assessed regularly (e.g., every 6 months).The researcher records changes in plaque levels, gum health, or caries.
This is a longitudinal study because the same participants are observed over time to track changes.

Quantitative Research: Correlational



- Correlational research is a type of quantitative study used to find out if there is a relationship between two or more variables without changing or controlling them.
- It does not prove cause and effect, only shows whether the variables are linked.
- The relationship can be positive, negative, or no correlation.
- Helps in predicting outcomes based on associations.
- Often uses statistical tools to measure the strength and direction of the relationship.

Quantitative Research: Correlational



- Example??????

A researcher wants to study the relationship between frequency of sugary snack intake and number of dental caries in children. They collect data on how often each child eats sugary snacks. They also record how many carious teeth each child has. Then, they analyze if children who eat more sugary snacks also have more caries.

This is correlational research because it looks for a connection between two naturally occurring variables — without altering them.

Quantitative Research: Survey

- A survey is a method of collecting information by asking people questions through questionnaires or interviews
- Can be written or oral (online, paper, or face-to-face).
- Involves a large group of people (sample).
- Can include closed-ended questions (yes/no, multiple choice)
- Surveys help gather data about opinions, behaviors, knowledge, or experiences

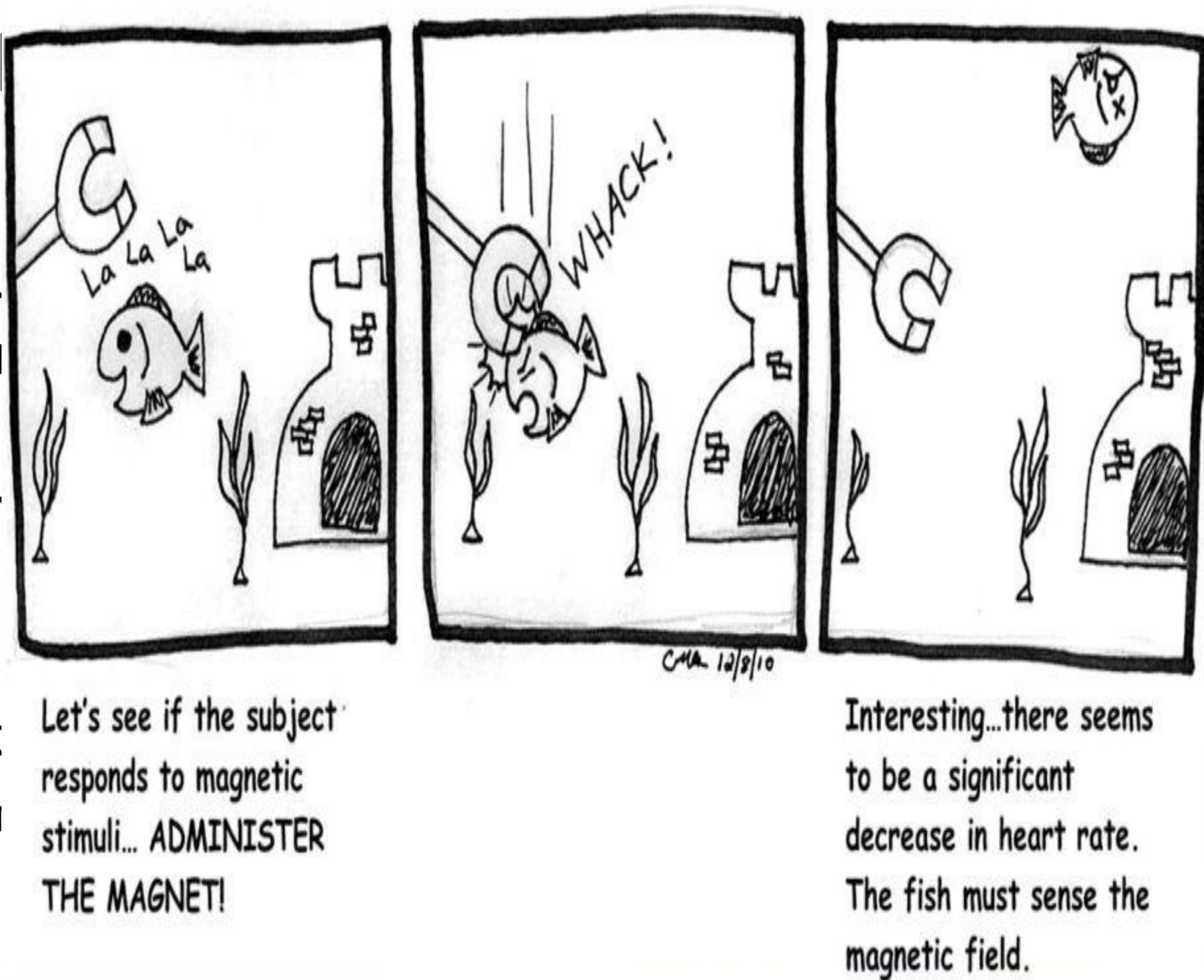
Quantitative Research: Survey

- Example??????

A researcher wants to know how often dental students brush their teeth and what toothpaste brands they use.
A questionnaire is given to 200 students asking about: Brushing frequency, Type of toothbrush, Brand of toothpaste,Use of mouthwash, etc. The researcher then analyzes the responses to identify trends.
This is survey research because data is collected directly from people by asking questions.

Quantitative Research: Quasi Experimental

- A quasi-experimental stu
  cause-and-effect relatio
  using existing classes or
- It is similar to an experir
  treatment).
- Less controlled and less
- Useful when randomizat
- Tries to compare outcon
  between two groups.



Quantitative Research: Quasi Experimental

- Example??????

A dental researcher wants to study the effectiveness of a new fluoride varnish in reducing dental caries.
One school uses the new fluoride varnish (intervention group).Another similar school continues regular dental care (comparison group). After 6 months, both groups are examined for caries development.
Since the schools were not randomly assigned, this is a quasi-experimental study — the researcher introduces a treatment but doesn't control all variables.

Quantitative Research: Experimental

- Experimental researc
  researcher actively m
  independent variable
  variable (the depende
- Participants are rand
  control groups
- It is used to test caus
  controlled conditions
- High level of control h
  validity.

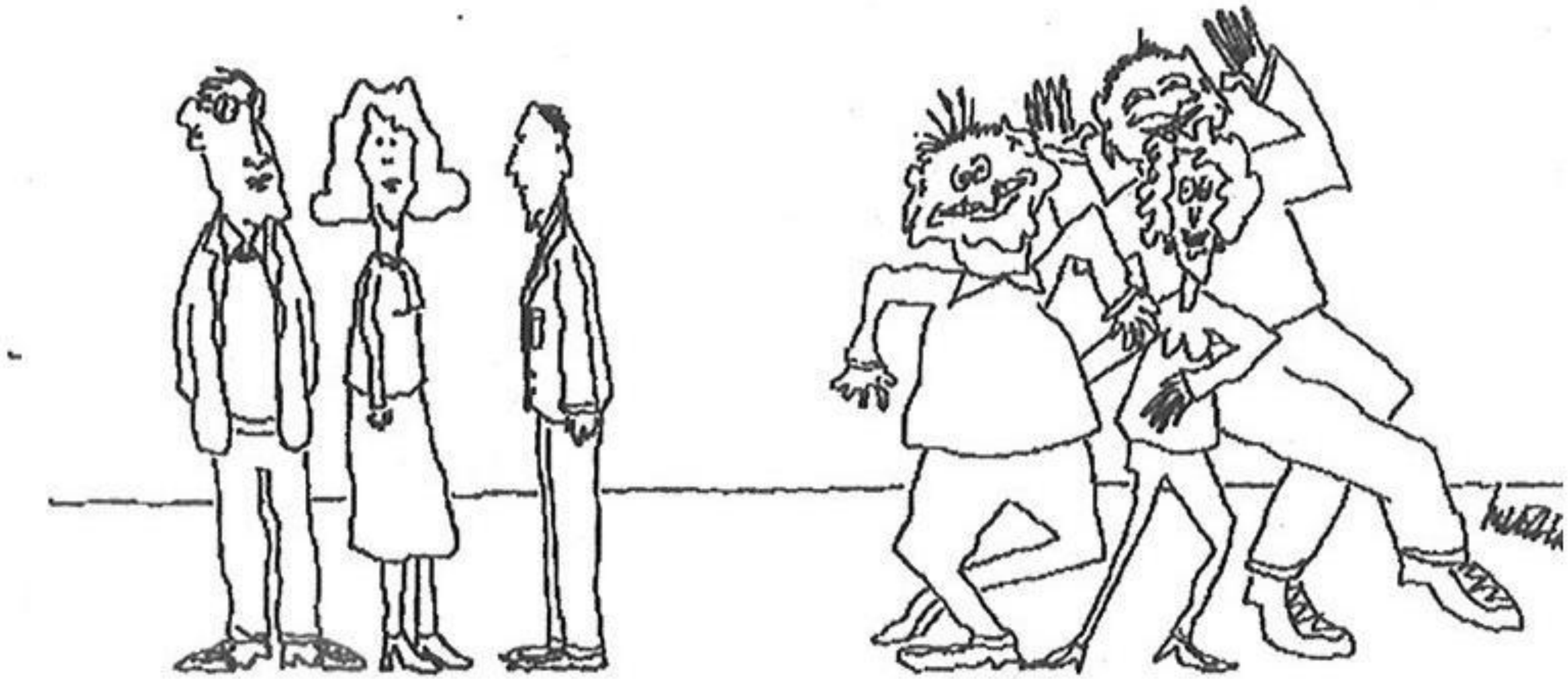
"There's a flaw in your experimental design.
All the mice are scorpios."

Quantitative Research: Experimental

"There's a flaw in your experimental design. All the mice are scorpios."

- Example??????

A dental researcher wants to test whether a new toothpaste reduces plaque better than a regular toothpaste.
100 participants are randomly assigned into two groups: Group A uses the new toothpaste. Group B uses a standard toothpaste. After 4 weeks, plaque levels are measured and compared.
This is experimental research because: The researcher manipulates the type of toothpaste. Participants are randomly assigned to groups. The effect (plaque reduction) is measured and compared.



CONTROL GROUP   OUT OF CONTROL GROUP.

- Snapshot
- Cross-sectional

- Looking backward
- Ex-post facto

- Link only
- Correlation

- Tracks changes
- Longitudinal

- Randomization
- Experimental

INDEPENDENT VS DEPENDENT VARIABLES

INDEPENDENT

An independent variable is the factor in an experiment that the researcher intentionally manipulates or changes to observe its effect. It's considered the cause or reason for an observed effect. Any changes in the dependent variable are hypothesized to be directly caused by variations in the independent variable.

DEPENDENT

The dependent variable is what the researcher measures to see if it changes as a result of the manipulation of the independent variable. It represents the outcome or effect in an experiment. Changes in the dependent variable are considered to be dependent on variations in the independent variable.

1- Fluoride concentration in tooth paste 2- Enamel mineralization level	**1- Independent 2- Dependent**
1- Time to straighten teeth 2- Type of bracket (Metal vs. Ceramic)	**1-Dependent 2- Independent**
1- Brushing technique 2- Gingival bleeding index	**1- Independent 2- Dependent**
1- Polishing speed 2- Surface roughness of enamel	**1- Independent 2- Dependent**
1- Risk of dental caries 2- Consumption of sugary drinks	**1-Dependent 2- Independent**

What is the primary characteristic of quantitative research?

A. It uses artistic interpretation
B. It involves numerical data
C. It focuses only on opinions
D. It avoids structured methodologies

B

Which type of quantitative research collects data at a single point in time?

A. Longitudinal

B. Cross-sectional

C. Experimental

D. Correlational

B

Which study design best establishes a cause-and-effect relationship?

A. Ex post facto
B. Cross-sectional
C. Experimental
D. Correlational

C

Which of the following is not a feature of survey research?

A. Use of interviews
B. Manipulation of variables
C. Use of questionnaires
D. Collection of opinions

B

In which design are participants randomly assigned to groups?

A. Quasi-experimental

B. Ex post facto

C. Experimental

D. Correlational

C

What distinguishes quasi-experimental from true experimental design?

A. No data collection

B. No hypothesis

C. Lack of random assignment

D. No intervention

C

Which method would a researcher use to analyze disease patterns over years?

A. Cross-sectional

B. Longitudinal

C. Correlational

D. Survey

B

Which research design involves looking backward in time?

A. Ex post facto
B. Experimental
C. Longitudinal
D. D. Survey

A

Correlational studies are best used to:

A. Establish causality
B. Compare treatments
C. Identify relationships between variables
D. Develop interventions

C

What is the first step in the research process?

A. Drawing conclusions
B. Reviewing literature
C. Sharing results
D. Identifying a problem or question

D

Which type of data is central to quantitative research?

A. Verbal
B. Observational
C. Numerical
D. Symbolic

C

Research Question:

Does regular use of fluoride toothpaste reduce the incidence of dental caries in school-aged children over a 6-month period?

Q: What is the independent variable in this study?

A: Use of fluoride toothpaste.

Q: What is the dependent variable?

A: Incidence of dental caries.

Q: What type of quantitative research design is most suitable for this question?

A: Experimental or quasi-experimental.

Q: How can researchers measure the incidence of dental caries?

A: Through clinical dental examinations before and after the 6-month period.

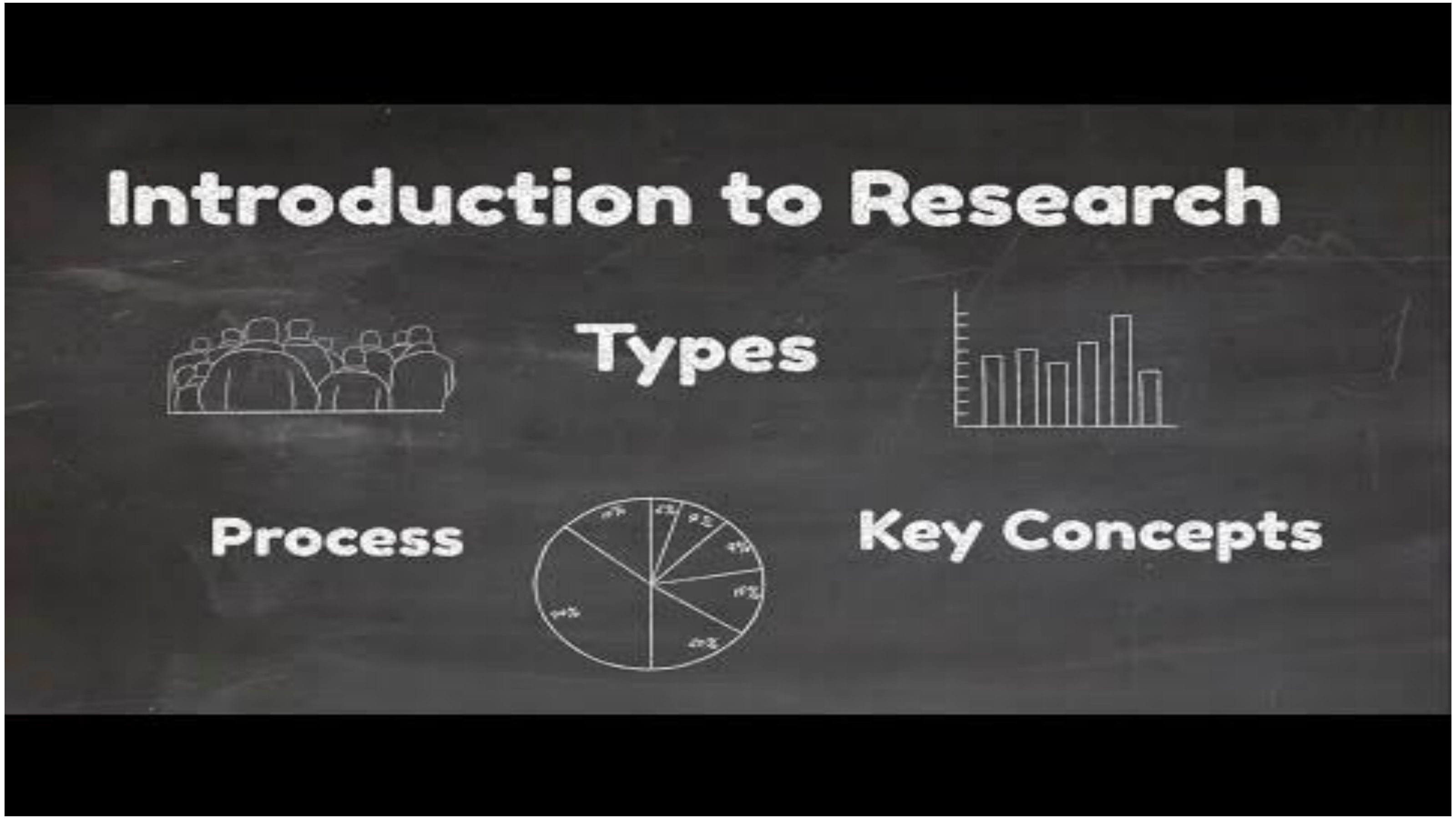
Q: Why is a control group necessary in this study?

A: To compare the effects of fluoride toothpaste with a non-fluoride alternative.

Q: How can bias be minimized in such a study?

A: By random assignment of participants and blinding.

Additional Read




Thank
You So
Much


END OF PRESENTATION
NO QUESTIONS, PLEASE